

mdstats FINAL-GPU1 Workstation Qualification Runbook

mdstats development

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DOWNSTREAM FINAL-GPU1 HANDOFF

This runbook describes the release-pinned, downstream FINAL-GPU1 handoff. It consumes a current P6 fresh-final-production publication and qualifies accelerator execution and release performance; it is not part of the current campaign lifecycle and does not provide target-size or model-selection authority. Replacement downstream qualification is a successor product obligation, not a P6 implementation shortcut.

1. Inputs

Prepare one workstation directory containing:

- the exact release source archive being qualified;
- the locked mace-mh-1.model and mace-mpa-0-medium.model files;
- the target/replay data required by the release-matched campaign;
- the project offline dependency bundle; and
- a writable qualification root.

Do not replace either foundation model. The tool verifies both model SHA-256 identities against the locked mdstats constants.

2. Runtime prerequisite

Activate the CUDA environment intended to authorize the release. It must contain the release-matched MACE/e3nn stack plus the CuEquivariance core, Torch frontend, and CUDA operations package appropriate to the installed PyTorch/CUDA runtime.

Missing accelerator capability is a failed/deferred runtime, not permission to alter scientific settings or silently fall back inside the same qualification record.

3. Verify and unpack the release

Verify the source archive with the checksum manifest supplied with the final bundle, then unpack it without editing source. The exact archive SHA-256 is part of every handoff registration.

Example:

```
sha256sum -c VERIFY_BUNDLE_SHA256.txt
mkdir -p work/source
unzip -q <release-archive.zip> -d work/source
cd work/source/<release-directory>
```

4. Downstream handoff preflight

```
python tools/run_mlff_final_gpu_qualification.py preflight \
  --mh1-model ../../inputs/mace-mh-1.model \
  --mpa0-model ../../inputs/mace-mpa-0-medium.model \
  --release-archive ../../<release-archive.zip> \
  --output ../../final-gpu1/preflight.json
```

This release-pinned preflight is the v11 handoff contract. A ready_for_final_gpu_execution result requires the locked models, a readable release archive, CUDA availability, and a positive CUEQ-DEP1 runtime freeze.

5. Initialize one immutable handoff root

```
python tools/run_mlff_final_gpu_qualification.py init \
  --root ../../final-gpu1/run-001 \
```

```
--mh1-model ../../../../inputs/mace-mh-1.model \
--mpa0-model ../../../../inputs/mace-mpa-0-medium.model \
--release-archive ../../../../<release-archive.zip>
```

The current policy creates a 16-item matrix: 8 must_pass, 6 measure_only, and 2 optional gates. Do not re-run init on a populated root. Replacement evidence belongs in a new run root.

6. Capture CUEQ-DEP1 once

```
mkdir -p ../../../../final-gpu1/run-001/raw
python tools/capture_mlff_cueq_depl_runtime.py \
--supplied-artifact ../../../../inputs/mace-mh-1.model \
--supplied-artifact ../../../../inputs/mace-mpa-0-medium.model \
--output ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json

python tools/run_mlff_final_gpu_qualification.py record \
--root ../../../../final-gpu1/run-001 \
--gate CUEQ_DEP1_RUNTIME_FREEZE \
--evidence ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json \
--disposition auto
```

All runtime-bound evidence must authenticate to this exact runtime digest.

7. Execute the current matrix

7.1 Must-pass gates

Gate ID	Required evidence
CUEQ_DEP1_RUNTIME_FREEZE	positive release-authorizing CuEq/CUDA runtime freeze
E3NN_BASELINE_COMPLETE_CAMPAIGN	complete optimized e3nn production-representative baseline
SIZE_FIDELITY1_EXHAUSTIVE_CALIBRATION	release qualification of the current size-screen calibration policy
PERF_P2R_WHOLE_FUNNEL_GPU_PERFORMANCE	whole-funnel GPU/control-plane performance with restart/decision equivalence
VRAM1_PERF_P4_ACCELERATOR_MEMORY_THROUGHPUT	VRAM/headroom/OOM/backoff and bounded-pipeline throughput evidence
CUEQ_PHASE1_TRAINING_ONLY_QUALIFICATION	paired e3nn/CuEq training qualification on one frozen runtime
REPLAY_UNIFY1_GPU_PSEUDOLABEL_EXECUTION	release-authoritative replay pseudolabel execution and cache/restart evidence
PERF_CERT1_END_TO_END_CERTIFICATION	end-to-end certification against the authoritative e3nn baseline

7.2 Measure-only gates

Register evidence for:

- PREC3_REAL_CUEQ_ACTIVATION
- MH1_ACCEL1_CUEQ_NUMERICAL_PARITY
- MH1_DATA6_1_CUEQ_DESCRIPTOR_SELECTION_PARITY
- MH1_TRAIN1_CUEQ_TRAINING_REALIZATION
- MH1_CERT1_GENERATED_DEFAULT_CUEQ_MATRIX
- PERF_P5_ACCELERATOR_PERSISTENCE_REUSE

These measurements must be complete even if a particular optimization remains disabled.

7.3 Optional gates

- CUEQ_PHASE2_SELECTED_HEAD_SOURCE_EXECUTION_OPTIONAL
- MH1_DEPLOY1_MLIAP_EXPORT_AND_LAMMPS_RUN0

8. Consume the current campaign publication

If a new campaign must be produced for this handoff, use the current staged CLI. Do not bypass target-size selection or held-out boundaries:

```
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> doctor
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> prepare
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> select-target-size --auto
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> cross-validate
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> train-production
```

The campaign itself owns the configured target-size ladder and fidelity path:

```
pi_train -> configured candidate ladder
          -> optional diagnostic screen(n1/M1 -> n2/M2 -> n3/M3) -> recommended N
          -> operator's provisional design
          -> cross-validate admission freezes N_selected,
              T_selected = pi_train[:N_selected], and both role horizons
          -> post-selection cross-validation on exactly T_selected
          -> fresh final production on the complete T_selected
```

select-target-size --auto adopts the diagnostic's recommendation; substitute select-target-size <N> to choose a qualified candidate explicitly. Either way, nothing is frozen until cross-validate admits it. FINAL-GPU1 must not create rescue sizes, migrate old target ladders, or alter a frozen target selection.

9. Register evidence

For every matrix item, use the handoff tool rather than editing the manifest:

```
python tools/run_mlff_final_gpu_qualification.py record \
  --root ../../../../final-gpu1/run-001 \
  --gate <GATE_ID> \
  --evidence ../../../../final-gpu1/run-001/raw/<evidence.json> \
  --disposition auto
```

The registration binds the evidence bytes, schema/content digest where available, release archive, and CuEq runtime digest where required.

10. Verify integrity before reduction

```
python tools/run_mlff_final_gpu_qualification.py verify \
  --root ../../../../final-gpu1/run-001
```

Any source/model/evidence mutation, policy/matrix drift, path escape, missing runtime binding, or inconsistent producer status fails closed.

11. Reduce FINAL-GPU1

```
python tools/run_mlff_final_gpu_qualification.py reduce \
  --root ../../../../final-gpu1/run-001 \
  --runtime ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json \
  --perf-cert1 ../../../../final-gpu1/run-001/raw/perf_cert1_qualification.json \
  --output ../../../../final-gpu1/run-001/FINAL_GPU1_QUALIFICATION.json
```

A positive record requires every must-pass gate to pass, every measure-only gate to be complete, handoff integrity to pass, and all runtime/release/model identities to agree.

A pass may recommend an accelerator profile but does not directly change generated defaults. The handoff record keeps generated-default authorization false.

12. Retired target-size migration workflow

Do **not** run retired migration, rescue, or target-data activation commands. Those belonged to historical campaign generations and are not prerequisites for the current campaign. Current campaigns reject retired derived target-size state before reuse and rebuild the target-size authority from the current P1/P2/P3 owners. The downstream handoff may reject a frozen final publication, but it may not select a different seed, checkpoint, member, or target size.